

Còc phng phòp tng hp thịnh phn C s v Chi tit trong bìi toàn Image Fusion

Bìi cnh

Trong còc kin trũc image fusion ãa trũn phón rĩ ãa tng (multi-scale decomposition — MSD) hoc ãa trũn hc sũu (nh CDDFuse [1]), nh ngun c tũch thịnh hai thịnh phn:

- **Thịnh phn c s** (*base / low-frequency*, kỳ hiu f^B) — biu ãin cu trũc tng th, tng phn chm, òt thay ãi gia còc modality.
- **Thịnh phn chi tit** (*detail / high-frequency*, kỳ hiu f^D) — biu ãin texture, biĩn, c trng riĩng tng modality.

Gi $a = f_V^B$ (hoc f_V^D) lĩ feature t modality VIS/MRI vĩ $b = f_I^B$ (hoc f_I^D) lĩ feature t modality IR/CT/PET/SPECT. Mi thịnh phn c tng hp bng mt *fusion rule* $R(a, b)$. Theo Liu et al. [2], quy trũnh MSD-based fusion gm 3 bc:

1. Phón rĩ: $I \rightarrow \{f^B, f^D\}$
2. Òp dng fusion rule: $f_F^B = R_B(f_V^B, f_I^B)$ vĩ $f_F^D = R_D(f_V^D, f_I^D)$
3. Tng hp ngc: $\{f_F^B, f_F^D\} \rightarrow I_F$

Tĩ liu nĩy lit kũ còc phng phòp R_B vĩ R_D ph bin, kũm cũng thc chi tit vĩ phng phòp mĩ CDDFuse [1] s dng.

Phng phòp CDDFuse gc s dng

CDDFuse [1] ãĩng **phũp cng n gin** (ng nht vĩ *trung bĩnh* sau LayerNorm) cho **c Base vĩ Detail**:

$$f_F^B = \text{BaseFuseLayer}(f_V^B + f_I^B), \quad f_F^D = \text{DetailFuseLayer}(f_V^D + f_I^D)$$

Trong ú BaseFuseLayer lĩ mt khi Transformer (LayerNorm + Attention + MLP) vĩ DetailFuseLayer lĩ mt khi INN (Invertible Neural Network). Phũp cng óy tn dng **shared encoder** (cũng tham s cho c hai modality, nõn channel cũng ng ngha) vĩ **decomposition loss** (ũp Base tng quan cao, Detail tng quan thp) hot ng ỹĩng.

1 Phng phòp tng hp thịnh phn C s

Mc tiĩu: bo toĩn cu trũc gĩi phu chung, trũnh artifact biĩn. Vĩ base lĩ vĩĩng tng quan cao gia còc modality, còc phng phòp thiĩn v *trung bĩnh cú trng s*.

1.1 Trung bình / Phép cộng (Average / Sum)

Cũng thế:

$$R(a, b) = \frac{1}{2}(a + b) \quad \text{hoc} \quad R(a, b) = a + b \quad (1)$$

Là baseline n gin nht, không cú tham s. Phép cộng vị trung bình ch khác nhau 1 h s 2, s b hp th bi LayerNorm/learnable scale khi refinement phòa sau [3].

⇒ CDDFuse [1] s đng phng phòp nịy.

1.2 Phng phòp Max (Choose-max)

Cũng thế:

$$R(a, b) = \begin{cases} a, & \text{nu } |a| > |b| \\ b, & \text{ngc li} \end{cases} \quad (2)$$

Ph bin cho high-frequency hn, him khi đúng cho base [4]. Cú th góy artifact biễn do "nhỹ"gia hai modality.

1.3 L1-norm weighted average

Activity measure: $A(a)_{i,j} = \|a_{i,j}\|_1 = \sum_c |a_c(i, j)|$ (tng L1 theo channel). Trng s:

$$w_a(i, j) = \frac{A(a)_{i,j}}{A(a)_{i,j} + A(b)_{i,j} + \epsilon}, \quad R(a, b) = w_a \cdot a + (1 - w_a) \cdot b \quad (3)$$

Trng s t l vì "tòch cc"ca feature. Modality nịo cú magnitude ln hn pixel ú s chim t trng cao hn. S đng trong DenseFuse [5].

1.4 Visual Saliency Map (VSM) weighted

Bn ni bt ti pixel p :

$$S(I)_p = \sum_{q \in N(p)} |I_p - I_q| \cdot w(I_q) \quad (4)$$

Trong ú $N(p)$ là ca s lớn cn ca p , $w(I_q)$ là trng s (thng histogram-based). Trng s fusion:

$$w_a = \frac{S(a)}{S(a) + S(b) + \epsilon}, \quad R(a, b) = w_a \cdot a + (1 - w_a) \cdot b \quad (5)$$

Vúng tng phn cao (mch màu, khi u, biễn xng) c u tiổn. Ph bin trong fusion y t [6, 7].

1.5 Local Energy weighted

Nng lng cc b:

$$E(a)_p = \sum_{q \in N(p)} a_q^2 \quad (6)$$

Trng s t l vì nng lng:

$$w_a = \frac{E(a)}{E(a) + E(b) + \epsilon}, \quad R(a, b) = w_a \cdot a + (1 - w_a) \cdot b \quad (7)$$

o cũng sut tồn hiu trong ca s $N(p)$ kờch thc $W \times W$ (thng $W = 3$ hoc 5). S đng trong nhiu s NSCT/NSST [2].

1.6 Local Variance / Entropy weighted

Phng sai cc b:

$$\sigma_a^2(p) = \frac{1}{|N(p)|} \sum_{q \in N(p)} (a_q - \mu_a(p))^2, \quad \mu_a(p) = \frac{1}{|N(p)|} \sum_{q \in N(p)} a_q \quad (8)$$

Entropy cc b (vi phón phi p_k ca bins trong ca s):

$$H_a(p) = - \sum_k p_k(a, N(p)) \log p_k(a, N(p)) \quad (9)$$

Trng s: $w_a = \sigma_a^2 / (\sigma_a^2 + \sigma_b^2 + \epsilon)$ hoc tng t vi entropy. o mc "ng" / "phong phũ thũng tin"[2].

1.7 Sparse Representation

Hc dictionary $D \in \mathbb{R}^{n \times K}$; biu din tha ca patch a vi b :

$$\alpha_a = \arg \min_{\alpha} \|\alpha\|_1 \text{ s.t. } \|a - D\alpha\|_2 \leq \delta \quad (10)$$

Fuse sparse code (vò d: $\alpha_F = \alpha_a + \alpha_b$ hoc max-rule), ri tòi to:

$$R(a, b) = D \cdot \alpha_F \quad (11)$$

Cú c s lý thuyt tt, sparse code cú ý ngha ng ngha rủ rịng [8]. Nhc im: optimization L1 chm, cn dictionary cht lng.

1.8 Adaptive Gating (Optimization-based)

Gate hc end-to-end qua loss:

$$g = \sigma(\text{Conv}_{1 \times 1}([a; b])) \in [0, 1]^{H \times W \times C}, \quad R(a, b) = g \odot a + (1 - g) \odot b \quad (12)$$

Trong ú $[a; b]$ li concat theo channel, σ li sigmoid, $\odot$ li phõp nhón Hadamard (per-pixel, per-channel). Trng s g hc theo tng pixel + tng channel [9, 10].

óy li phng phõp òn xut (CDDFuse-AG — Adaptive Gating).

1.9 Pretrained Deep Features

Trõch c trng t VGG19 [11] hoc ResNet [12]:

$$\phi(a) = \text{VGG19}_{\text{relu1_1}}(a), \quad \phi(b) = \text{VGG19}_{\text{relu1_1}}(b) \quad (13)$$

Tõnh activity t activation map (thng L1-norm hoc L2-norm):

$$A_a = \|\phi(a)\|_1, \quad w_a = \frac{A_a}{A_a + A_b + \epsilon} \quad (14)$$

Tn dng kin thc t ImageNet ònh giò quan trng vúng nh (*transfer learning*).

Bng 1: Cc phng phỏp tng hp thnh phn c s.

Phng phỏp	Cũng thc / Ỗ tng	Tham kho
Sum (paper CDDFuse)	$R(a, b) = a + b$ — khũng tham s	[3, 1]
Average	$R(a, b) = (a + b)/2$ — tng ng Sum sau LayerNorm	[3]
L1-norm	$w_a = \ a\ _1 / (\ a\ _1 + \ b\ _1)$	[5]
VSM	$w_a \propto \sum_N I_p - I_q w(I_q)$	[6, 7]
Local Energy	$w_a \propto \sum_N a^2$ — cũng sut cc b	[2]
Local Var/Entropy	$w_a \propto \sigma_a^2$ hoc H_a	[2]
Sparse Rep.	Fuse trũn sparse code $\alpha = \arg \min \ \alpha\ _1$ s.t. $\ a - D\alpha\ _2 \leq \delta$	[8]
Adaptive Gating	$g = \sigma(\text{Conv}_{1 \times 1}([a; b]))$ hc qua loss	[9]
Pretrained DL	$\phi = \text{VGG19}(\cdot)$, $w_a \propto \ \phi(a)\ _1$	[11, 12]

1.10 Bng tng hp

2 Phng phỏp tng hp thnh phn Chi tit

Mc tiũu: gi li tũn hũu mnh nht t mi modality (texture mũ mm MRI, rim xng CT). Vĩ detail li vũng tng quan thp (mi modality mang c trng riũng), cc phng phỏp thiũn v *la chn* (selection) hoc *kũch hot* (activity-based).

CDDFuse dũng cho Detail

CDDFuse cng dũng **phũp cng** cho Detail: $f_F^D = \text{DetailFuseLayer}(f_V^D + f_I^D)$, vi **DetailFuseLayer** li khi INN. ỹ li *la chn khũng chũn* so vi trũn thng (trũn thng thng dũng max-abs cho detail). Lý do CDDFuse vn dũng c: decomposition loss ỹ cc_D ≈ 0 (Detail ca hai modality khũng tng quan), nõn phũp cng = phũp *tng hp b tr* (mi modality ỹng gũp phn riũng) ch khũng phi pha loũng.

2.1 Choose-max-abs (trũn thng)

$$R(a, b) = \begin{cases} a, & \text{nu } |a| > |b| \\ b, & \text{ngc li} \end{cases} \quad (15)$$

Quy tc c in nht cho high-frequency [4]. Chn h s cú biũn ln nht, gi nh "ln hn = cha thũng tin nhiu hn".

Soft version (differentiable):

$$w_a = \sigma\left(\frac{|a| - |b|}{\tau}\right), \quad R(a, b) = w_a \cdot a + (1 - w_a) \cdot b \quad (16)$$

vi τ li tham s nhit; khi $\tau \rightarrow 0$ tin ti hard max-abs.

2.2 Spatial Frequency (SF)

Row Frequency vĩ Column Frequency:

$$RF(a)_p = a_p - a_{p-1}, \quad CF(a)_p = a_p - a_{p-W} \quad (17)$$

$$SF(a)_p = \sqrt{RF(a)_p^2 + CF(a)_p^2} \quad (18)$$

Trng s:

$$w_a(p) = \frac{SF(a)_p}{SF(a)_p + SF(b)_p + \epsilon} \quad (19)$$

o " ng"ca texture. Ph bin trong multi-focus fusion [13].

2.3 Sum-Modified-Laplacian (SML)

Modified Laplacian ti pixel p :

$$ML(a)_p = |2a_p - a_{p-1} - a_{p+1}| + |2a_p - a_{p-W} - a_{p+W}| \quad (20)$$

Sum-Modified-Laplacian trong ca s $N(p)$:

$$SML(a)_p = \sum_{q \in N(p)} ML(a)_q \quad (21)$$

Trng s: $w_a = SML(a)/(SML(a) + SML(b) + \epsilon)$. o sc nôt (sharpness) [14].

2.4 Local Energy/Variance per-channel

Tng t thịnh phn c s (Mc 1), nhng tòngh *per-channel* (gi tòngh chn lc cao ca detail):

$$E_c(a)_p = \sum_{q \in N(p)} a_c(q)^2, \quad w_a(c, p) = \frac{E_c(a)_p}{E_c(a)_p + E_c(b)_p + \epsilon} \quad (22)$$

2.5 Pulse-Coupled Neural Network (PCNN)

Mng neural gai [15] mĩ phng v nĩu th giĩc mĩu. Mi pixel lĩ 1 neuron vi 5 thịnh phn:

Feeding input: $F_{ij}[n] = a_{ij}$

Linking input: $L_{ij}[n] = V_L \sum_{kl \in N(ij)} W_{ijkl} Y_{kl}[n-1]$

Internal activity: $U_{ij}[n] = F_{ij}(1 + \beta L_{ij})$

Dynamic threshold: $\theta_{ij}[n] = \theta_{ij}[n-1]e^{-\alpha_\theta} + V_\theta Y_{ij}[n-1]$

Pulse output:

$$Y_{ij}[n] = \begin{cases} 1, & U_{ij} > \theta_{ij} \\ 0, & \text{ngc li} \end{cases} \quad (23)$$

Chy T vùng lp ($T \sim 100-200$), m *fire frequency*:

$$\text{Fire}(a)_{ij} = \sum_{n=1}^T Y_{ij}[n] \quad (24)$$

Trng s fusion:

$$w_a = \frac{\text{Fire}(a)}{\text{Fire}(a) + \text{Fire}(b) + \epsilon} \quad (25)$$

Bin th quan trng:

- **PA-PCNN** (Parameter-Adaptive PCNN) [16]: tham s $\beta, \alpha_\theta, V_\theta$ t thit lp t c trng nh, khĩng cn tune th cũng.
- **Dual-Channel PCNN (DC-PCNN)** [17]: hai PCNN gĩĩp ỹi cho hai modality, link 2 chiu.

2.6 Saliency-guided gate

Tính bn gradient (Sobel hoc tng t):

$$G(a)_p = \sqrt{(\partial_x a)^2 + (\partial_y a)^2} \Big|_p \quad (26)$$

Trng s:

$$w_a = \frac{G(a)}{G(a) + G(b) + \epsilon}, \quad R(a, b) = w_a \cdot a + (1 - w_a) \cdot b \quad (27)$$

ng b vi supervision signal nu loss function cng da trón saliency (nh Saliency-guided Pixel loss) [18].

2.7 Sparse coefficient max

Tng t thnh phn c s; vi detail thng đứng max-rule trón sparse code:

$$\alpha_F(k) = \begin{cases} \alpha_a(k), & \text{nu } |\alpha_a(k)| > |\alpha_b(k)| \\ \alpha_b(k), & \text{ngc li} \end{cases} \quad (28)$$

[8].

2.8 Adaptive Gating

Ging Mc 1, òp dng cho detail vi gate hc riõng:

$$g_D = \sigma(\text{Conv}_{1 \times 1}([a_D; b_D])), \quad R(a_D, b_D) = g_D \odot a_D + (1 - g_D) \odot b_D \quad (29)$$

2.9 Bng tng hp

Bng 2: Cóc phng phòp tng hp thnh phn chi tit.

Phng phòp	Cũng thc / Ý tng	Tham kho
Sum (paper CDDFuse)	$R(a, b) = a + b$ qua INN block	[1]
Choose-max-abs	a nu $ a > b $, ngc li b	[4]
Spatial Frequency	$w_a \propto \sqrt{RF^2 + CF^2}$	[13]
SML	$w_a \propto \sum_N 2a_p - a_{p-1} - a_{p+1} + 2a_p - a_{p-W} - a_{p+W} $	[14]
Local Energy/Var	$w_a \propto \sum_N a_c^2$ per-channel	[2]
PCNN / PA-PCNN	$w_a \propto \sum_{n=1}^T Y[n]$ (fire frequency)	[15, 16]
Saliency-guided	$w_a \propto \sqrt{(\partial_x a)^2 + (\partial_y a)^2}$	[18]
Sparse Rep.	$\alpha_F = \text{max-rule trón sparse code}$	[8]
Adaptive Gating	$g_D = \sigma(\text{Conv}([a; b]))$ hc qua loss	[9]

3 Nhãn tốt chung

- **Base vs Detail đúng rule khác nhau:** Liu et al. [2] không nh base thng đúng *trung bình / weighted average*, detail thng đúng *max-abs* hoc *activity-based*.
- **CDDFuse [1] lệ ngoại lệ:** đúng *phộp cng n gín cho c hai*. S thịnh cũng ca thit k nỳ da trộn shared encoder + decomposition loss (õp Base correlated, Detail uncorrelated), khin phộp cng hot ng ững mị khững cn rule phc tp.
- **Xu hng hin i:** Adaptive Gating (PP6) vị Pretrained DL (PP7) ang chim u th nh kh nng hc theo d liu, kt hp vi transfer learning t ImageNet hoc checkpoint pretrained.
- **Trade-off:** Rule rigid (max, average) n gín, khững cn train, nhng kỗm linh hot. Rule hc c (gating, deep features) mnh hn nhng ùi hi dataset, compute, vị training stability.

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